

Dynamic Pricing Optimization for Centric Brands

— Technical Brief

This document is a structured walk-through of how dynamic price optimization can be modeled and solved in a realistic retail scenario, with constraints that Centric Brands would care about. It is designed to demonstrate both the business relevance and the mathematical rigor you would expect in a technical interview.

1. Business Context

Centric Brands manages multiple categories (e.g., denim, hoodies, accessories) across channels. Key challenges include:

- **Inventory risk:** avoid leftover stock while preserving margin.
- **Channel constraints:** MAP (minimum advertised price), parity with retail partners, clearance floors.
- **Stability constraints:** avoid large week-to-week price swings.
- **Margin constraints:** ensure each sale covers at least a minimum markup.

The goal is to set a price path across the season (8–12 weeks, for example) that maximizes total profit, while respecting brand and channel constraints.

1.a Scenario Assumptions (used in the numerical example)

- **Product (primary):** Women's Denim Jacket (owned brand).
- **Secondary (optional):** Kids Hoodie (owned brand) — used when showing multi-SKU or cannibalization.
- **Channel:** Direct-to-Consumer (D2C) e-commerce on Centric-managed sites. (Wholesale/ marketplaces are referenced via parity constraints when included.)
- **Horizon & cadence:** 8 weeks, weekly price decisions (can be daily in production; weekly is used here for clarity and to match seasonality/promos).
- **Final price grids (discrete):**
 - **Denim (A):** {\$79, \$89, \$99, \$109, \$119, \$129, \$139}
 - **Hoodie (B):** {\$39, \$44, \$49, \$54, \$59, \$64, \$69, \$74}
- **Costs (COGS):** Denim \$38; Hoodie \$18.
- **Floors:** Denim MAP \$99; Clearance floor \$79 (Weeks 5 & 8).
- **Hoodie MAP:** \$49; Clearance floor \$39 (Weeks 5 & 8).
- **Price-move caps:** Denim \$10 per week; Hoodie \$8 per week.
- **Initial inventories:** Denim 400 units; Hoodie 300 units.
- **End-of-season leftover cap:** $\leq 5\%$ of initial inventory.
- **Competitor price paths:** Included as exogenous features in demand; parity constraints can be turned on/off (off in the base example; on in the parity section).

These assumptions drive the demand curves and all DP optimization shown below.

2. Demand Modeling Choices

The core of pricing optimization is a demand model: how sales volume responds to price.

(a) Constant Elasticity (Log-Log)

- $D(p) = A \cdot (\text{MSRP}/p)^\varepsilon$, with $\varepsilon < 0$ (e.g., -1.2 to -2.0).
- **Pros:** interpretable, widely used, easy to estimate (linear regression in logs).
- **Cons:** assumes elasticity is constant across the price band.

(b) Log-Linear (Semi-Log)

- $\ln D = \alpha + \beta p \Rightarrow D(p) = Ae^{\beta p}$.
- **Pros:** absolute price has constant impact (useful at lower price bands).
- **Cons:** may under- or over-estimate elasticity away from calibration point.

(c) Hybrid / Piecewise

- Splines or tree-based models to capture non-linearities.
- Useful with rich transaction data, but harder to embed in optimization.

Why Log-Log here?

- Simple, interpretable, scale-free.
- Elasticity estimates are directly actionable ("1% price cut $\rightarrow$ 1.4% more sales").
- Good fit for apparel categories with broad price bands.

3. Constraints to Include

We embed realistic brand rules:

1. **Price grid:** discrete candidate prices (e.g., {79, 89, 99, 109, 119, 129}).
2. **MAP / clearance floors:**
3. Normal weeks: $p_t \geq \text{MAP}_t$.
4. Clearance weeks: $p_t \geq \text{ClearanceFloor}_t$.
5. **Price move cap:** $|p_t - p_{t-1}| \leq \Delta_{\max}$.
6. **Margin rule:** $p_t - c \geq m_{\min}$.
7. **Inventory flow:** $I_t = I_{t-1} - s_t$.
8. **Leftover cap:** $I_T \leq L_{\text{target}}$ (e.g., $\leq 5\%$ of initial stock).

4. Optimization Formulation

We want to maximize total profit:

$$\max_{\{p_t\}} \sum_{t=1}^T (p_t - c) s_t$$

subject to the constraints above.

Because decisions today affect inventory tomorrow, we solve it as a dynamic program.

Bellman Equation (core)

$$V(t, I, p_{t-1}) = \max_{p_t \in \text{Feasible}(t, p_{t-1})} \left[(p_t - c) \cdot s_t + V(t+1, I - s_t, p_t) \right],$$

with

$$s_t = \min\{I, \lfloor D_t(p_t) \rfloor\}, \quad V(T+1, I, \cdot) = \begin{cases} 0, & I \leq L_{\text{target}} \\ -\infty, & I > L_{\text{target}} \end{cases}$$

This recursively enforces the leftover constraint and bakes in all price feasibility rules (MAP/clearance, price-move cap, min margin, parity when enabled).

4.a Bellman in Action — Step-by-Step Numeric (mini 2-week demo)

Toy setup (Denim A) for intuition:

- **Horizon:** 2 weeks ($t = 1, 2$).
- **Start inventory ($t=1$):** $I_1 = 200$.
- **Cost:** $c = \$38$.
- **Price grid:** $\{99, 109, 129\}$.
- **Demand (log-log):** $D(p) = A(p/129)^\varepsilon$ with $A = 120$, $\varepsilon = -1.4$.
- **End leftover cap:** ≤ 10 units.

Terminal (Week 2): given p_2 , the value is $\max(p_2 - c) \cdot \min(I_2, \lfloor D(p_2) \rfloor)$ if $I_2 \leq 10$; else infeasible.

Week 1: for each p_1 , compute profit now + best Week-2 value.

- $p_1 = 129$: sales = 120, profit = 10,920; next inventory = 80; Week-2 value = 7,280 → **Total = 18,200**.
- $p_1 = 109$: sales = 144, profit = 10,224; next inventory = 56; Week-2 value = 5,096 → **Total = 15,320**.
- $p_1 = 99$: sales = 165, profit = 10,065; next inventory = 35; Week-2 value = 3,185 → **Total = 13,250**.

Decision: choose **129** at Week 1 because it maximizes profit now + future value.

In the full 8-week DP, the same recursion applies each week, with seasonality, marketing lifts, floors, move caps, and leftover cap encoded in the terminal value.

5. Numerical Example (8-week horizon)

Scenario (Denim A):

- **MSRP:** \$129; **Cost:** \$38; **Inventory:** 400; **Elasticity:** -1.4 .
- **Price grid:** {79, 89, 99, 109, 119, 129, 139}.
- **Floors:** MAP \$99, clearance \$79 (weeks 5, 8).
- **Move cap:** \$10.
- **Leftover cap:** 20 units.

Optimal 8-week price path (DP result):

- **Denim (A):** [129, 129, 119, 109, 99, 109, 109, 99]
- **Hoodie (B):** [69, 69, 64, 59, 54, 54, 49, 49]

Why this path?

- **Weeks 1–2:** hold premium prices (129/69) while demand is strong and inventory not at risk.
- **Weeks 3–4:** step down gradually within move caps to stimulate volume.
- **Week 5 (promo lift + clearance):** drop to 99/54 to unlock volume at peak demand.
- **Weeks 6–7:** stabilize prices (109/54→49) to balance margin vs. sell-through.
- **Week 8 (final clearance):** finish low (99/49) to ensure leftover $\leq 5\%$ of stock.

This plan respects all constraints (floors, move caps, margins, leftover cap) and achieves high sell-through with controlled markdowns.

6. Extensions

- **Stochastic demand:** multiply $D(p)$ by a lognormal shock; use robust DP (pessimistic quantile) for downside protection.
 - **Cannibalization:** model two SKUs jointly with cross-elasticities $D_A(p_A, p_B)$, $D_B(p_B, p_A)$.
 - **Reinforcement Learning:** replace DP with Q-learning/DQN when demand model is uncertain and you want the system to learn directly from interaction.
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7. Why This Matters for Centric Brands

- **Direct match to reality:** leftover caps, MAP rules, price-move caps, and clearance windows are exactly the constraints Centric faces.
 - **Actionable output:** the model produces a week-by-week price path.
 - **Flexibility:** can integrate parity with partners, stochastic demand, or RL exploration.
 - **Competitive edge:** turns pricing into a systematic, data-driven lever, not manual guesswork.
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8. Key Takeaways for Interview

- I understand the business constraints. MAP, clearance, move caps, margin floors, inventory are all included.
- I can write the optimization mathematically. Clear Bellman equation with constraints.
- I can do the numbers. Step-by-step numeric example and full 8-week plan.
- I know alternatives. Log-log vs log-linear vs hybrid, and why log-log is reasonable here.
- I see the future. Mentioned stochastic demand, cannibalization, and reinforcement learning as extensions.

This is the kind of system Centric Brands could deploy today to unlock incremental margin while respecting brand guardrails.

Appendix: Section Outline Snapshot

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